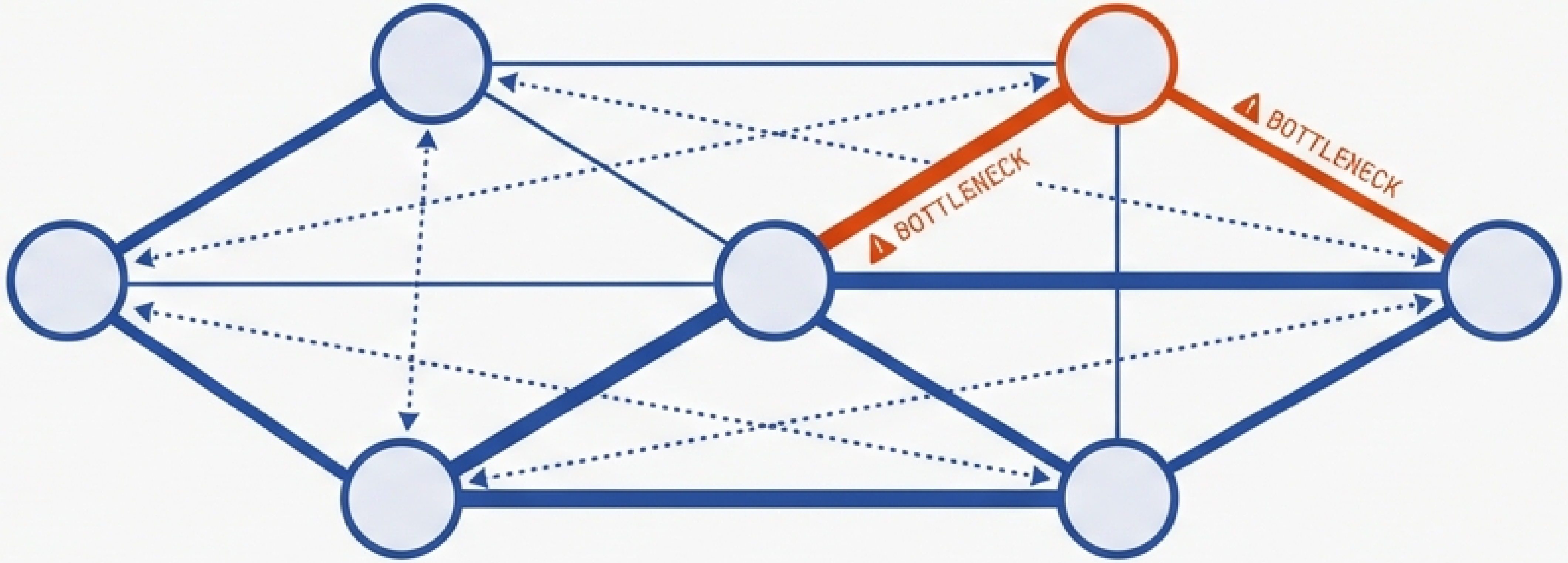
ISSUES-FS ROLE ARCHITECTURE

A STRATEGIC STRESS-TEST

Framework Analysis & Implementation Roadmap (v1.0)



DATE: 05 FEB 2026 STATUS: DRAFT CONTEXT: AGENT COORDINATION PROTOCOL

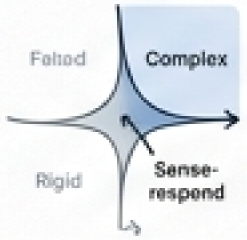
EXECUTIVE SUMMARY

The architecture is viable but carries high coordination risks.

STRATEGIC (WARDLEY)

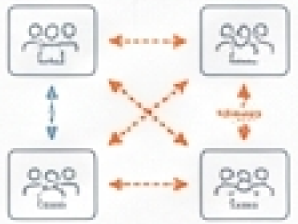
The Conductor role and State Machine are the unique differentiators (Genesis). They require disproportionate investment compared to commodity components like DevOps.

OPERATIONAL (CYNEFIN)



The system behaves as a ‘Complex’ domain. Rigid upfront specification will fail. Emergent design and probe-sense-respond tactics are required.

STRUCTURAL (SPOTIFY)



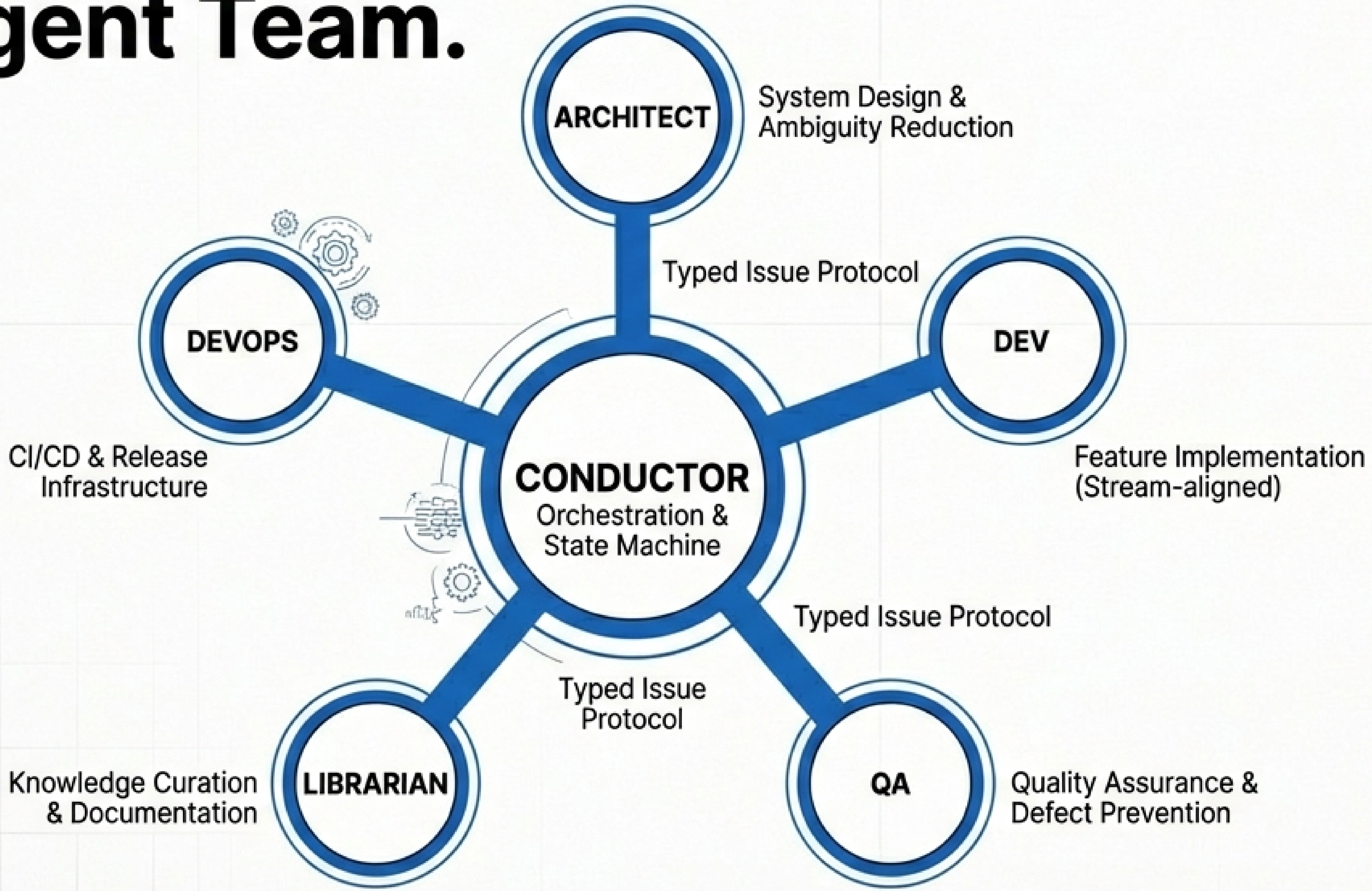
Critical shared conventions (‘Chapters’) are missing. Without a mechanism for cross-role standards, there is a significant risk of role drift and siloed knowledge.

PRIMARY RECOMMENDATION

Shift from a ‘Big Bang’ rollout to a ‘Safe-to-Fail’ probe involving only only three roles: Conductor + Dev + 1. Measure flow before scaling.

THE CONTEXT

The Ecosystem: A Hub-and-Spoke Agent Team.

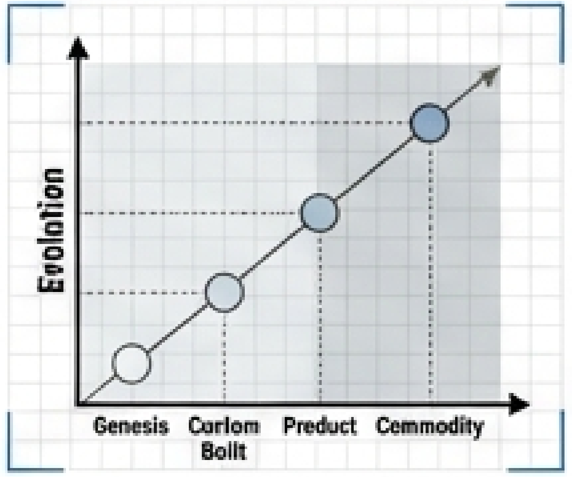


THE PROBLEM

Coordination of multi-agent systems is usually chaotic. This architecture attempts to solve it via a strict, typed-issue coordination protocol.

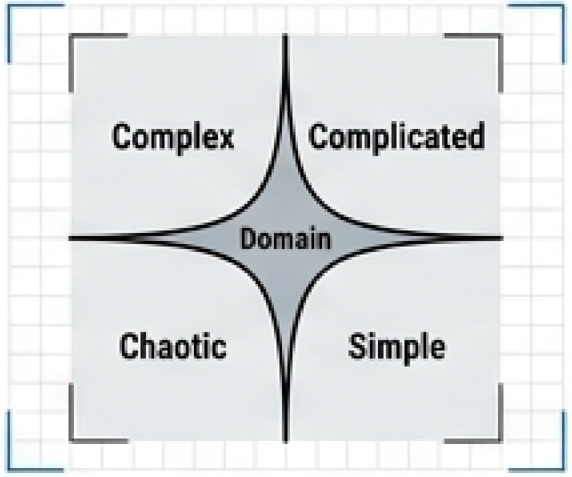
METHODOLOGY

Five lenses to identify structural weakness



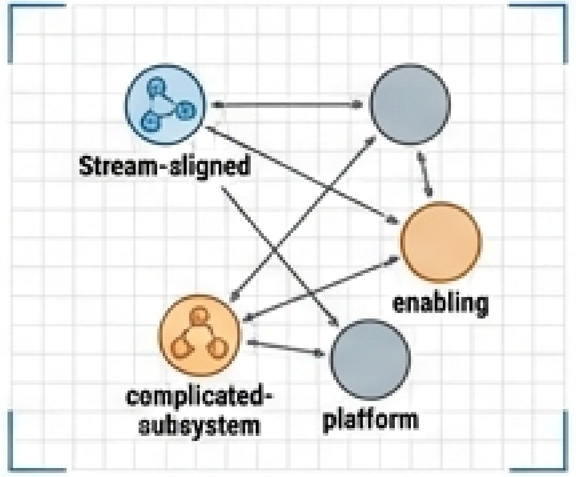
WARDLEY MAPS
Strategy & Evolution

Focus: Value Chain



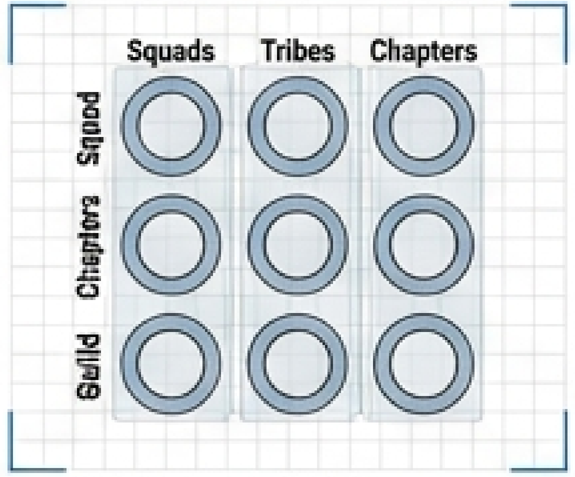
CYNEFIN
Complexity & Decisions

Focus: Uncertainty



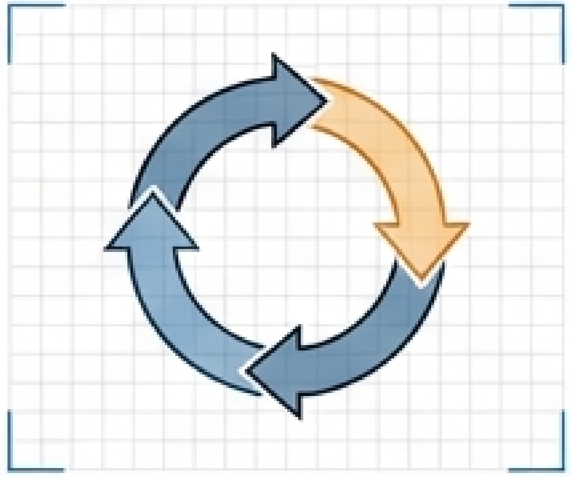
TEAM TOPOLOGIES
Interaction & Flow

Focus: Cognitive Load



SPOTIFY MODEL
Scalability & Autonomy

Focus: Structure



AMAZON
Customer-Focus & Metrics

Focus: Velocity

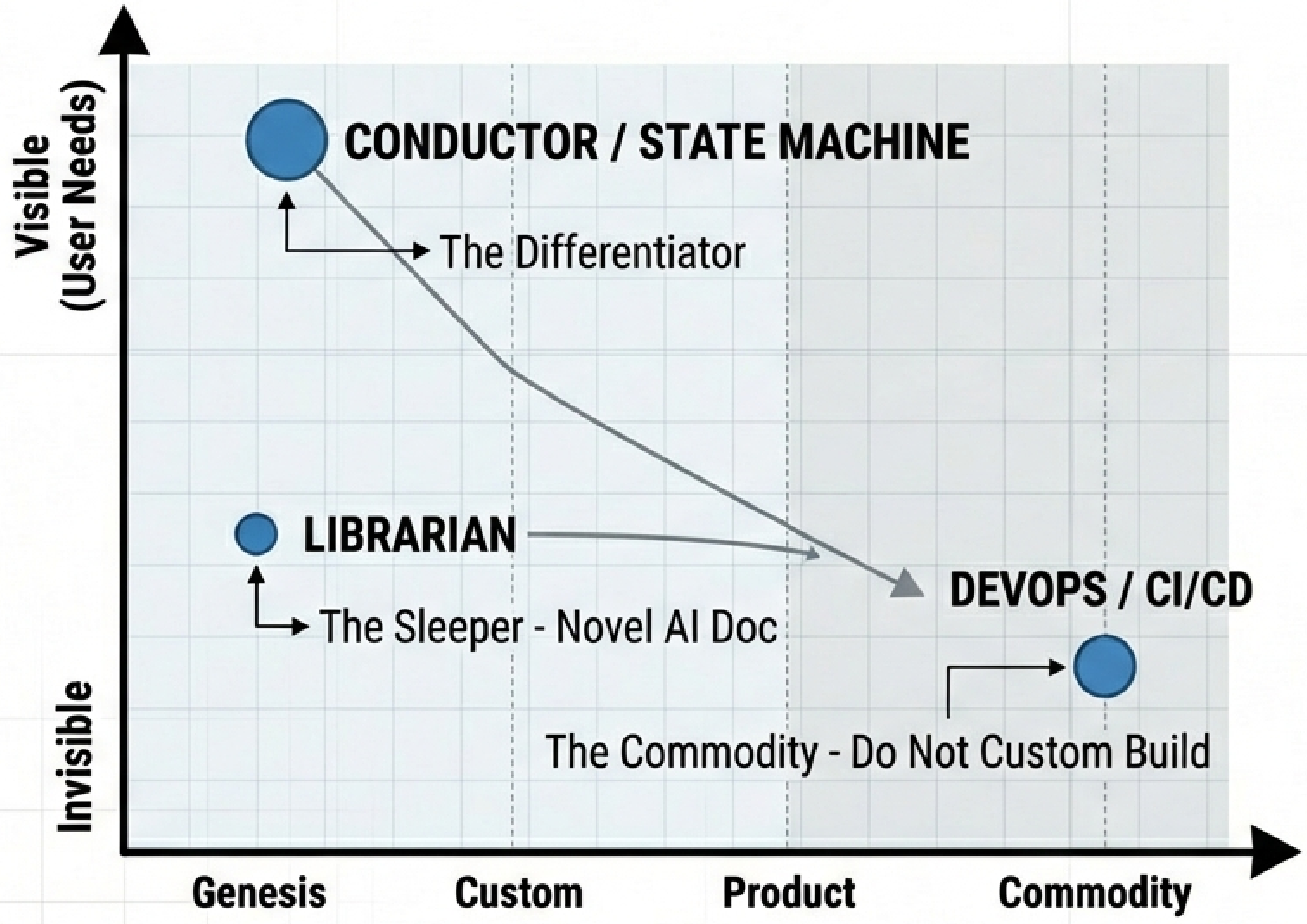
GOAL: To move from “Complicated” design to “Complex” reality without failure.

CLASSIFICATION

CLASSIFICATION

LENS 1: WARDLEY MAPS (STRATEGY)

Focus Innovation on Coordination; Commoditise DevOps.



KEY INSIGHTS

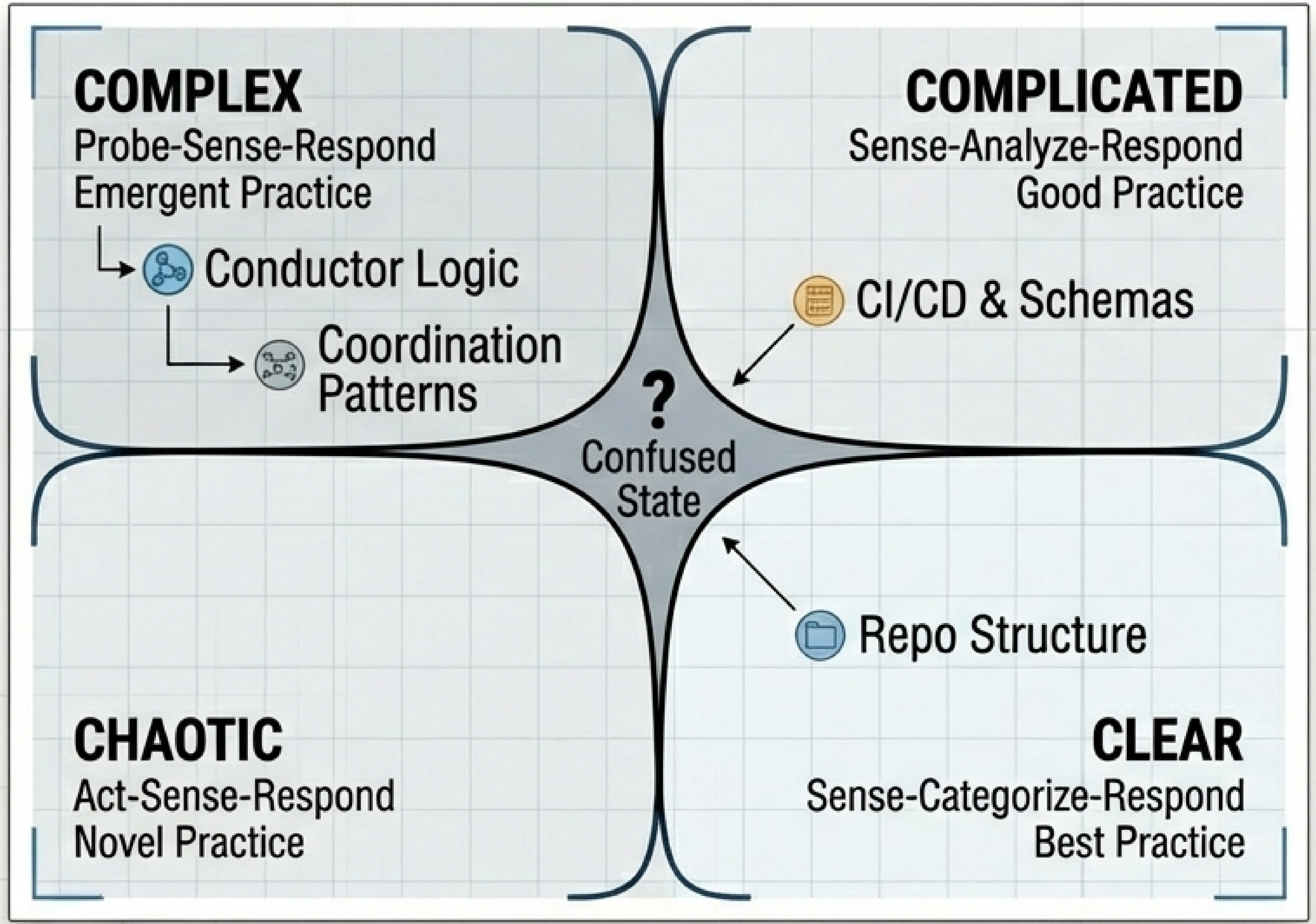
1. **THE DIFFERENTIATOR:** The **Typed-Issue State Machine** is the ‘secret sauce’. If this fails, the system fails.

2. **THE COMMODITY:** Do not build custom CI/CD tools. Use standard GitHub Actions.

3. **THE SLEEPER:** The **Librarian** role is more novel than it appears—no playbook exists for AI documentation agents.

LENS 2: CYNEFIN FRAMEWORK (COMPLEXITY)

Design for Emergence, Not Specification.



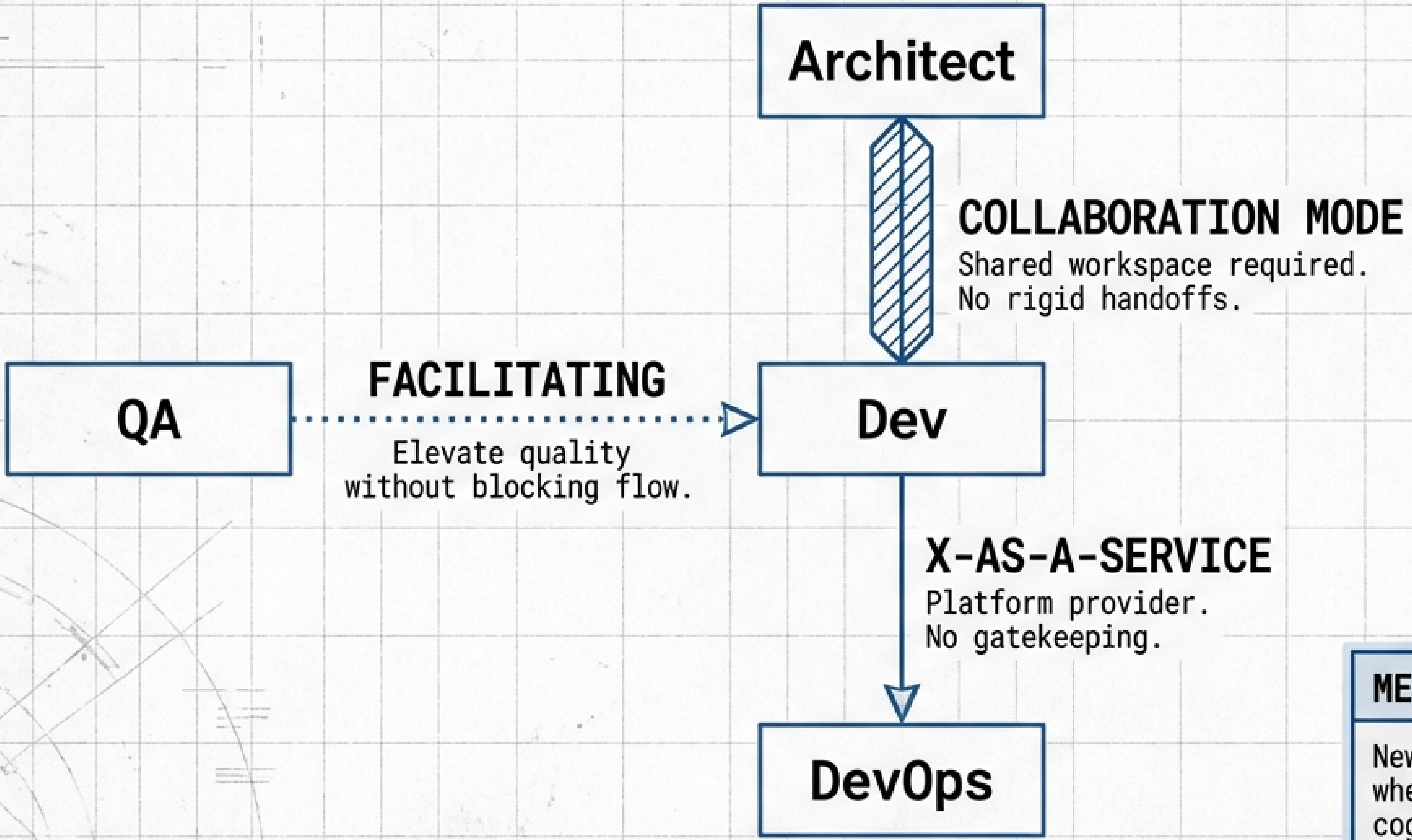
KEY INSIGHTS

- **Complex Domain:** How agents coordinate is Complex (cause and effect only seen in retrospect).
- **Probe-Sense-Respond:** We cannot fully specify the state machine upfront. We must launch, observe friction, and adapt.
- **Confused State:** Roles need an “escape hatch” protocol for unclassified blockers.

CLASSIFIED

REPORT NO. 2022-04-
CLASSIFIED

Explicit Interaction Modes Reduce Friction.



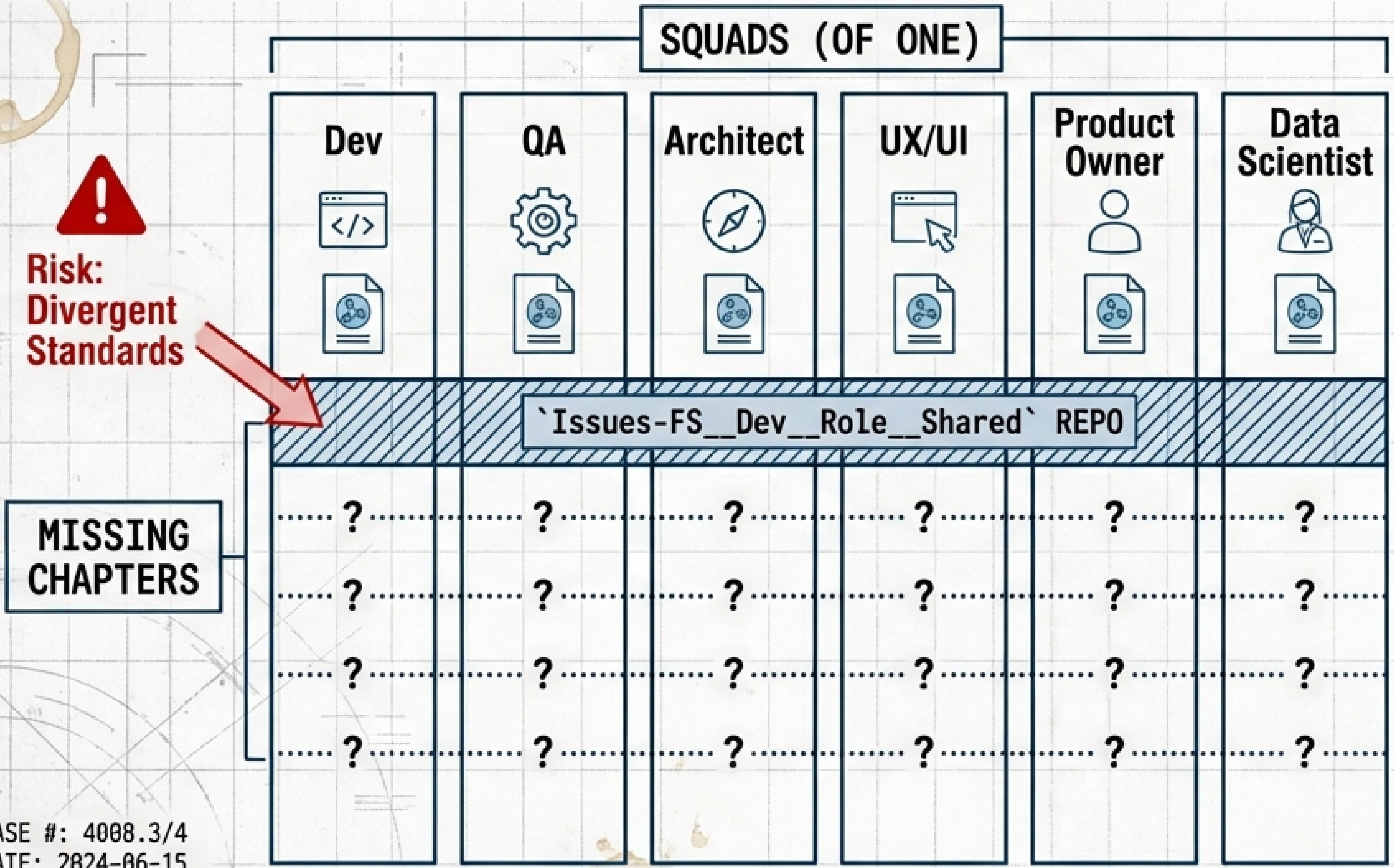
METRIC: COGNITIVE LOAD

New roles must be assessed on whether they reduce or increase the cognitive load on the Dev agent.

CLASSIFIED

REPORT NO. 2022-04-CLASSIFIED

Solos, Tribes and Missing Chapters.



KEY INSIGHTS

- **Squad of One:** The agent system is a single squad.
- **The Missing Link:** No mechanism for cross-role standards (e.g. Schemas).
- **Solution:** Create a shared repo to act as the 'Chapter'—a single source of truth.

METRIC: CROSS-ROLE ALIGNMENT

Success is measured by the adoption of shared standards and reduction in role-specific silos.

CASE #: 4008.3/4
DATE: 2024-06-15
STATUS: ACTIVE

CLASSIFIED

REPORT NO. 2022-04-
CLASSIFIED

Work Backwards from Flow Metrics.

Agent Overhead

Cost is linear to “handoff types”, not team size.

Working Backwards

Define success metrics NOW:

- Cycle time
- Defect escape rate
- Blocker resolution time.

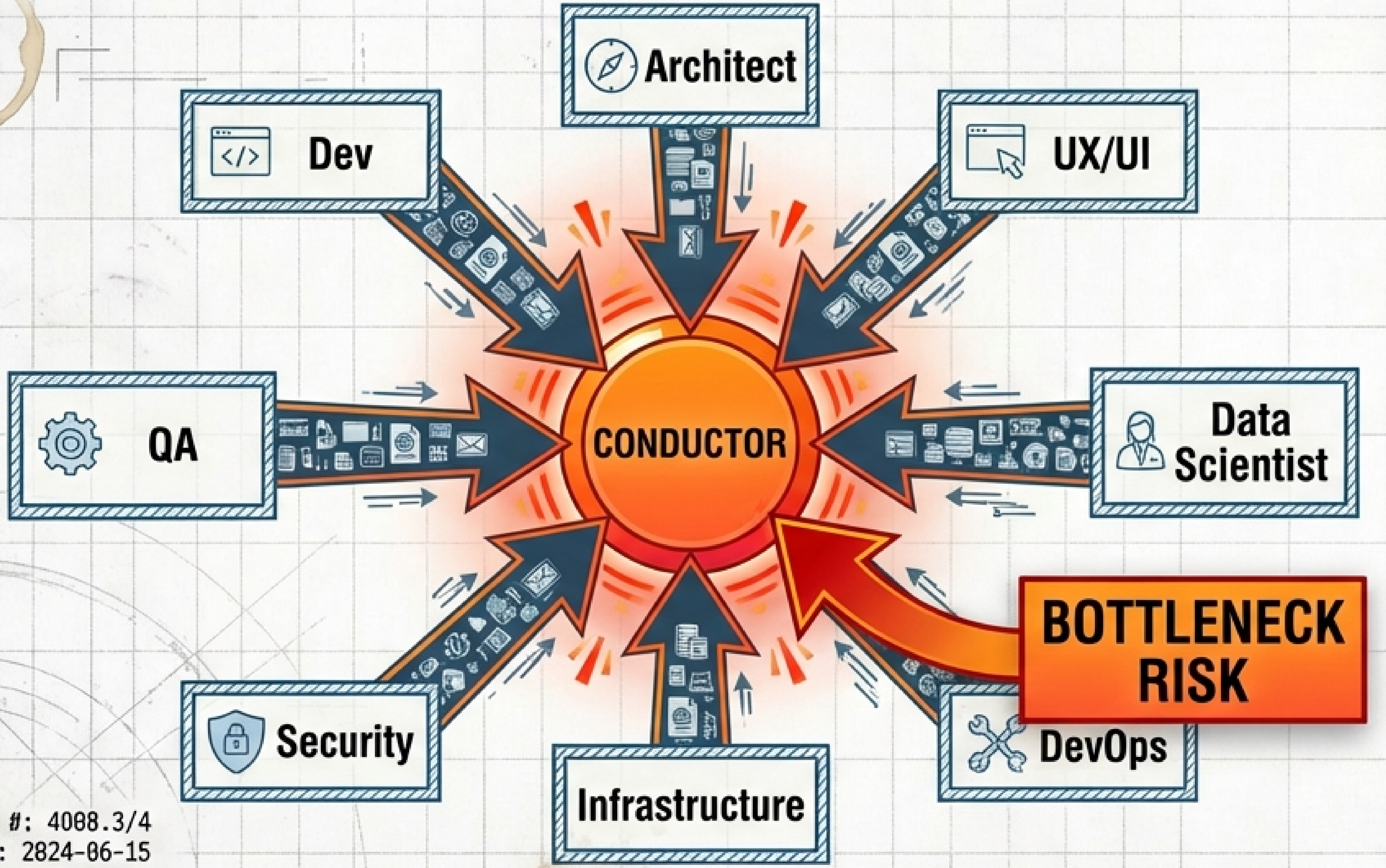
Blast Radius

If one role (e.g., QA) fails, the factory must not stop. Bypass protocols required.

CLASSIFIED

REPORT NO. 2022-04-
CLASSIFIED

The Conductor is the Critical Failure Point.



The Risk:

The Hub-and-Spoke model creates a quadratic complexity risk if every task goes through the centre.

The Mitigation:

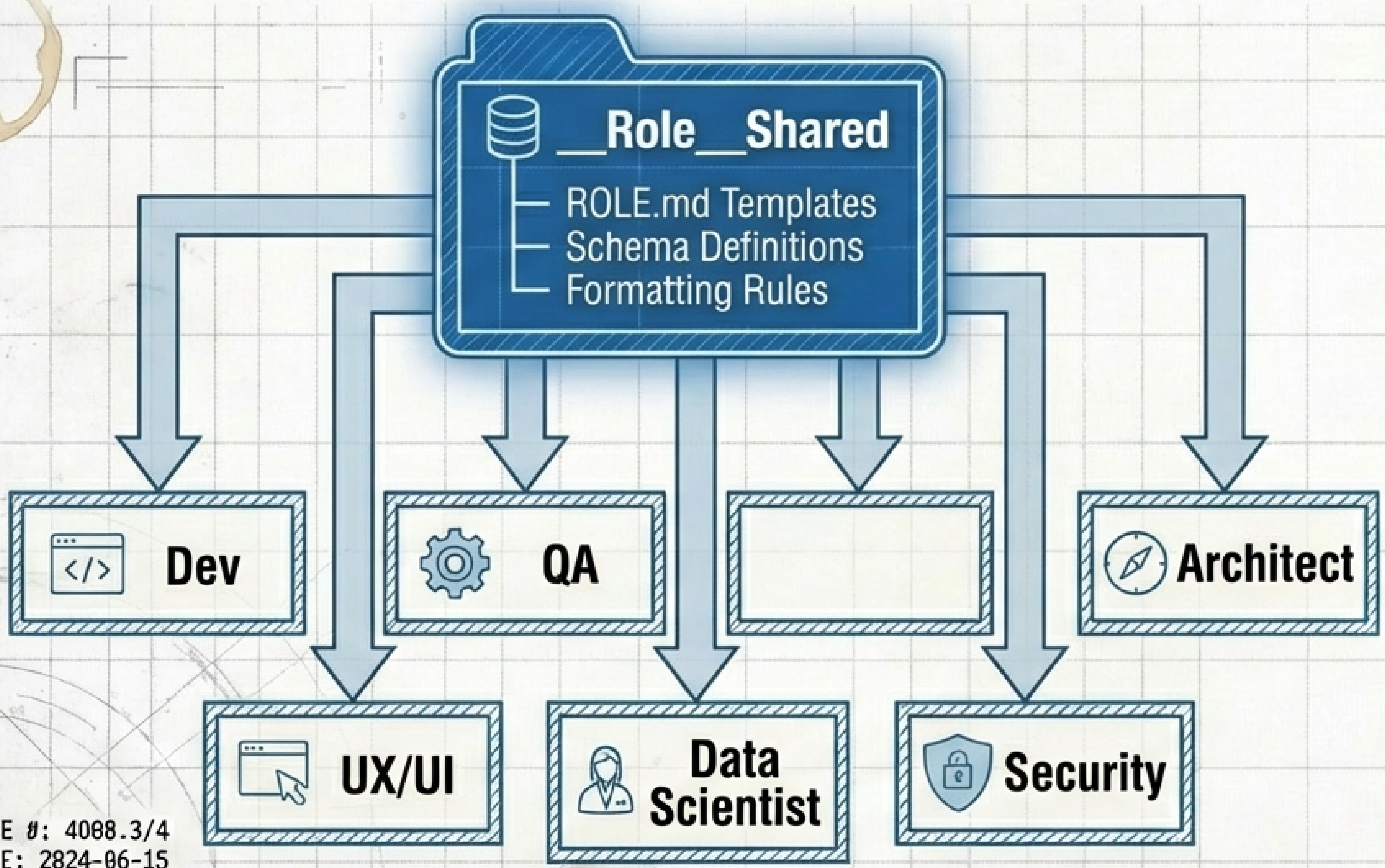
1. Flow Optimisation (manage flow, not tasks).
2. Bypass Protocols (allow direct Dev->DevOps interaction).
3. Retrospectives (self-analysis of latency).

CASE #: 4008.3/4
DATE: 2024-06-15
STATUS: CRITICAL

CLASSIFIED

REPORT NO. 2022-04-
CLASSIFIED

Preventing Knowledge Silos.



The Risk:

Roles are autonomous but isolated. Schemas will drift.

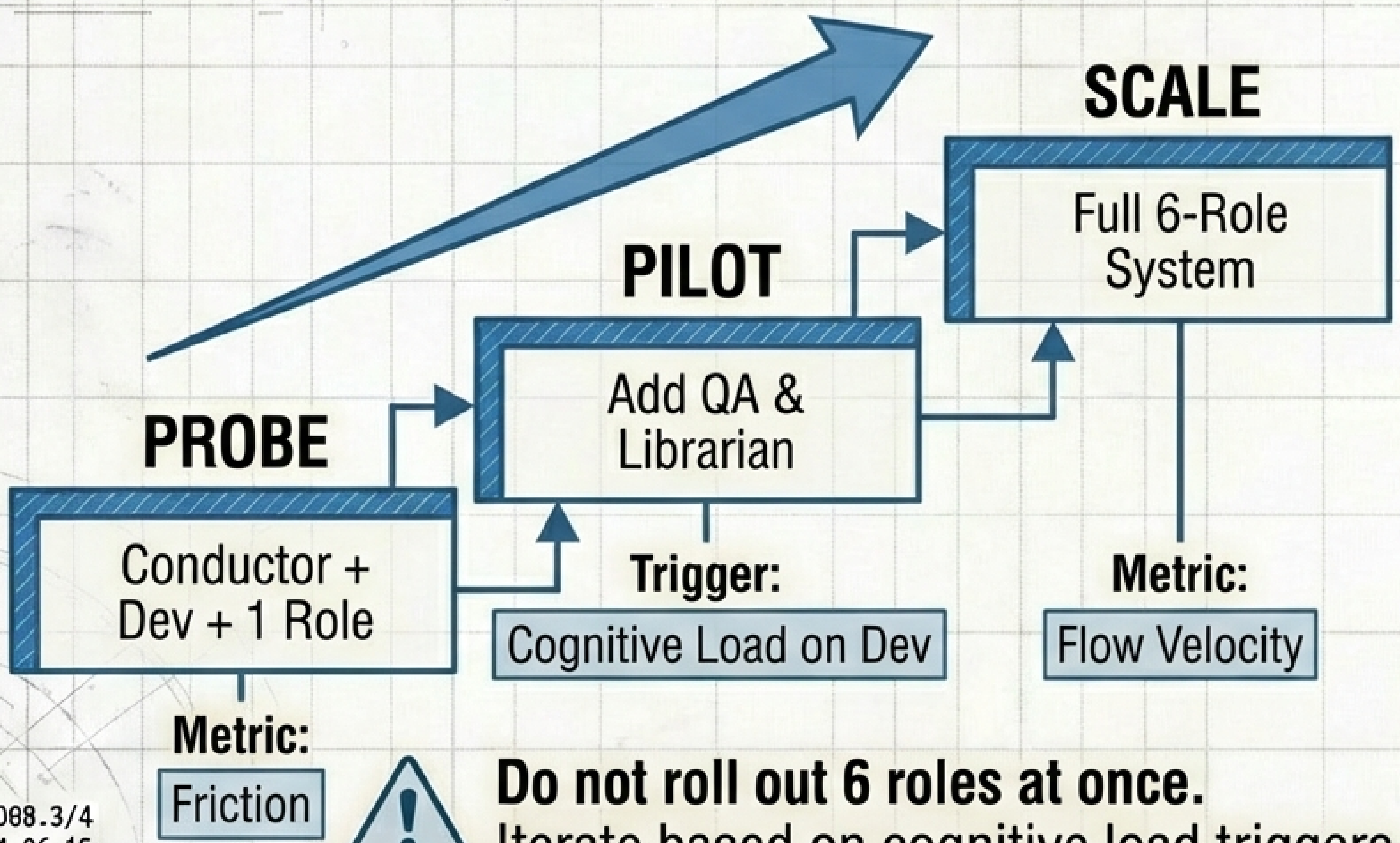
The Fix: A central Constitution repo.

Guilds: Periodic `Review_Request' issues where multiple roles review API designs together.

CASE #: 4008.3/4
DATE: 2824-06-15
STATUS: CRITICAL

CLASSIFIED

The ‘Safe-to-Fail’ Approach: Start Small, Learn Fast.



Do not roll out 6 roles at once.
Iterate based on cognitive load triggers.

The Risk:

Full 6-Role system are trigict a complexity tin the Lowe’ siide.

The Fix: A centratc with Constitution lewellard.

Guilds: Benodics ‘support to strongle’ job, nomentic, fatt, auttroach and complexler directors.

REPORT NO. 2022-04-
GLASSJFIED

CASE #: 4008.3/4
DATE: 2824-06-15
STATUS: CRITICAL

CLASSIFIED

REPORT NO. 2022-04-
CLASSIFIED

The Foundation.

01. METRIC DEFINITION [SOURCE: AMAZON]

- Define success metrics (Cycle time, Defect escape) before building a single agent.

02. STRATEGIC INVESTMENT [SOURCE: WARDLEY]

- Focus 80% of engineering effort on the Coordination Protocol (State Machine). This is the genesis component.

03. PROBING [SOURCE: CYNEFIN]

- Launch the ‘Safe-to-Fail’ probe.
  Do not build the full `ROLE.md` for all 6 roles yet.

CASE #: 4008.3/4
DATE: 2824-06-15
STATUS: CRITICAL

CLASSIFIED

REPORT NO. 2022-04-
CLASSIFIED

The Guardrails.

01. SHARED REPO [SOURCE: SPOTIFY]

- Create `__Role__Shared` for conventions immediately to prevent drift.

02. INTERACTION MODES [SOURCE: TEAM TOPOLOGIES]

- Explicitly define interaction types (Collaboration vs Service) in every `ROLE.md`.

03. CONDUCTOR SAFETY [SOURCE: AMAZON/CYNEFIN]

- Implement bottleneck monitoring and retrospective mechanisms for the Conductor.

04. BYPASS PROTOCOLS [SOURCE: AMAZON]

- Define failure paths—what happens if the Architect goes offline? Who authorises the bypass?

CASE #: 4008.3/4
DATE: 2824-06-15
STATUS: CRITICAL

CLASSIFIED

REPORT NO. 2022-04-

GLASSJFIED

Moving from ‘Draft’ to ‘Genesis’.

“The role architecture is not a static blueprint; it is a complex adaptive system. We must build it to learn.”

- The **Conductor** is the differentiator.
- Shared practices are the glue.
- **Flow** is the metric.

CASE #: 4088.3/4
DATE: 2824-06-15
STATUS: CRITICAL

Frameworks: Simon Wardley (Maps), Dave Snowden (Cynefin),
Skelton & Pais (Team Topologies), Spotify Labs, Amazon (Working Backwards).